

Comparative Evaluation of Cox Proportional Hazards and Random Survival Forest Models for Overall Survival Prediction in Breast Cancer

An Independent Computational Study Using the METABRIC Cohort

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Research focus: Survival analysis, breast cancer prognosis, machine learning, clinical prediction, model discrimination, calibration, and uncertainty quantification.

Study design: Retrospective computational analysis of a publicly available breast cancer cohort.

Primary comparison: Penalized Cox proportional hazards regression versus Random Survival Forest using an identical clinical predictor set and held-out test cohort.

Abstract

Background: Predicting overall survival in breast cancer is challenging because survival outcomes are affected by heterogeneous clinical characteristics and incomplete follow-up. Cox proportional hazards regression provides an established statistical framework for analyzing censored survival outcomes, whereas Random Survival Forests offer a flexible machine-learning alternative capable of representing nonlinear relationships and interactions.

Objective: To compare the discrimination, five-year prediction accuracy, and calibration of penalized Cox proportional hazards regression and Random Survival Forest models using clinical characteristics from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) cohort.

Methods: A retrospective computational study was conducted using 1,980 patients included in the final modeling cohort. Patients were divided into a training cohort of 1,583 and a held-out test cohort of 397. Both models used the same seven clinical predictors: age at diagnosis, tumor size, number of positive lymph nodes, tumor grade, estrogen receptor status, progesterone receptor status, and HER2 status. A penalized Cox model and a Random Survival Forest were fitted using the training cohort. Discrimination was evaluated using Harrell's concordance index. Five-year prediction accuracy was evaluated using the inverse probability of censoring weighted (IPCW) Brier score. Calibration was examined by comparing predicted five-year survival probabilities with Kaplan–Meier estimates across prediction-based patient groups. Paired bootstrap resampling with 2,000 iterations was used to estimate confidence intervals for differences in test-set performance.

Results: The Cox model achieved a test concordance index of 0.6548, compared with 0.6912 for the Random Survival Forest. The observed difference was 0.0364, with a 95% paired bootstrap confidence interval of 0.0089 to 0.0655. The five-year IPCW Brier scores were 0.1570 for Cox regression and 0.1451 for the Random Survival Forest. The difference in Brier scores (Cox minus Random Survival Forest) was 0.0119 (95% bootstrap confidence interval: 0.0023 to 0.0219). Both models showed differences between predicted and observed survival in some calibration groups.

Conclusion: The Random Survival Forest demonstrated higher discrimination and lower five-year prediction error than penalized Cox regression in the held-out METABRIC test cohort. These findings illustrate the potential value of flexible machine-learning approaches for clinical survival prediction. However, the results are limited to an internal evaluation of one retrospective cohort and require independent validation before clinical application.

Keywords: Breast cancer; METABRIC; overall survival; Cox proportional hazards; Random Survival Forest; survival prediction; concordance index; Brier score; calibration.

1 Introduction

1.1 Background and motivation

Breast cancer is a heterogeneous disease in which patients may experience substantially different clinical outcomes despite receiving the same diagnosis. Differences in patient characteristics, tumor burden, histological grade, receptor status, and other biological factors contribute to variation in disease progression and survival.

Estimating patient-level survival probabilities is therefore an important statistical and computational problem. Prognostic models may help researchers investigate associations between clinical characteristics and outcomes, identify patterns of risk, and evaluate the potential usefulness of different modeling approaches.

A fundamental difficulty in survival analysis is that the outcome is not completely observed for every patient. Some patients remain alive at the end of follow-up, while others are lost to follow-up before experiencing the event of interest. These observations are described as right-censored.

Conventional regression methods that ignore censoring may produce misleading results because they cannot distinguish an observed event from incomplete follow-up. Survival analysis methods address this problem by jointly considering event occurrence and the duration of observation.

The Cox proportional hazards model is among the most widely used approaches for analyzing time-to-event data. Its formulation relates covariates to the instantaneous event hazard without requiring specification of the baseline hazard function (Cox, 1972).

Although Cox regression is interpretable and statistically well established, its conventional linear predictor and proportional hazards assumption may limit its ability to represent complex relationships in heterogeneous clinical data.

Machine-learning approaches provide an alternative framework for survival prediction. Random Survival Forests extend ensemble tree methods to right-censored outcomes and can represent nonlinear associations and interactions without requiring the same parametric structure as conventional Cox regression (Ishwaran et al., 2008).

However, greater model flexibility does not automatically translate into improved prediction on previously unseen patients. More complex models may overfit training data, and their predicted survival probabilities may be poorly calibrated even when their risk rankings are informative.

Consequently, survival prediction models should be compared using multiple complementary evaluation criteria rather than relying on a single performance statistic.

1.2 Clinical prediction and model evaluation

Survival prediction involves several related but distinct objectives.

First, a model may be used to rank patients according to their relative risk of experiencing an event. This property is commonly referred to as discrimination.

Second, a model may estimate the probability that a patient survives beyond a clinically relevant time horizon. Evaluating these probabilities requires consideration of prediction error and censoring.

Third, predicted probabilities should correspond reasonably well to observed outcomes. This property is known as calibration.

A model may discriminate between higher-risk and lower-risk patients while systematically overestimating or underestimating their survival probabilities. Therefore, discrimination alone is insufficient to characterize the quality of an individual survival prediction model.

The present study evaluates both discrimination and five-year probabilistic prediction accuracy, with an additional grouped calibration analysis.

1.3 Research objective

The primary objective of this study was to compare a penalized Cox proportional hazards model with a Random Survival Forest for predicting overall survival in breast cancer using the METABRIC cohort.

The comparison was designed to address three research questions:

1. How do the two models compare in their ability to discriminate between patients with different observed survival outcomes?
2. How accurately do the models estimate five-year survival probabilities when right-censoring is taken into account?
3. How closely do the predicted five-year survival probabilities correspond to observed survival across patient groups?

A secondary objective was to quantify uncertainty in the observed differences between the models using paired bootstrap resampling of the held-out test cohort.

The study focuses on predictive performance rather than causal inference. Model associations and predictions are not interpreted as evidence that modifying a particular clinical characteristic would change an individual patient's survival.

2 Data and Study Design

2.1 Data source

The study used clinical and overall survival information from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) cohort.

METABRIC is a breast cancer research resource developed to investigate molecular and clinical heterogeneity. Its original research publications describe genomic and transcriptomic analyses of breast cancer and their relationship to clinical characteristics and outcomes (Curtis et al., 2012; Pereira et al., 2016).

The present analysis was restricted to a clinical predictor set and did not incorporate gene expression, genomic alterations, or molecular subtype features.

This restriction allowed the comparison to focus on the predictive behavior of two survival modeling approaches when provided with the same routinely interpretable clinical variables.

The analysis was retrospective and computational. No new patients were recruited, and no treatment decisions were made using the predictions generated in this study.

2.2 Study population

The processed survival cohort contained 1,981 patient records before the final modeling eligibility check.

One observation with a survival duration of zero months was excluded from the predictive modeling analysis.

The resulting modeling cohort comprised 1,980 patients.

An 80:20 training–test split was used, with stratification by observed event status and a fixed random seed of 42.

The training cohort contained 1,583 patients, and the held-out test cohort contained 397 patients.

The test cohort was reserved for final model evaluation and was not used to select the Random Survival Forest hyperparameters.

Table 1: Study population and data partitioning.

Characteristic	Number
Processed survival cohort	1,981
Excluded zero-duration observation	1
Final modeling cohort	1,980
Training cohort	1,583
Held-out test cohort	397
Training events	915
Test events	229
Training censored observations	668
Test censored observations	168

The event counts in Table 1 refer to observed events during the available follow-up period, not exclusively to deaths within the first five years.

2.3 Outcome definition

The primary outcome was overall survival, represented by two variables:

- **OS_MONTHS**: observed follow-up duration in months;
- **EVENT**: event indicator, with 1 representing an observed death and 0 representing a censored observation.

For patient i , the observed outcome was represented as

$$Y_i = (T_i, \delta_i), \quad (1)$$

where T_i denotes the observed follow-up time and δ_i indicates whether the event occurred.

A censored observation indicates that the event time was not observed within the available follow-up period. It does not imply that the patient would remain alive indefinitely.

The primary prediction horizon was 60 months, corresponding to five years.

2.4 Clinical predictors

Seven clinical variables were included in both predictive models.

Table 2: Clinical predictors included in both models.

Predictor	Representation
Age at diagnosis	Continuous numerical variable
Tumor size	Continuous numerical variable
Positive lymph nodes	Numerical count
Tumor grade	Categorical variable
Estrogen receptor status	Categorical variable
Progesterone receptor status	Categorical variable
HER2 status	Categorical variable

Age at diagnosis, tumor size, and positive lymph node count were treated as numerical predictors.

Tumor grade and receptor status were represented using categorical indicator variables.

After preprocessing, the seven original clinical predictors were represented by 12 model input columns.

Both models received the same processed feature matrices, thereby reducing the possibility that differences in their performance resulted from different predictor sets.

2.5 Data preprocessing

The preprocessing workflow included handling missing predictor values and encoding categorical variables.

Preprocessing parameters were estimated using the training cohort and then applied to the held-out test cohort.

The test cohort was not used to estimate imputation values or determine the categorical encoding structure.

The resulting matrices contained 1,583 training observations and 397 test observations, each with 12 processed features.

The same preprocessing output was used for Cox regression and the Random Survival Forest.

This common input representation allowed the subsequent comparison to focus on differences between the modeling algorithms.

3 Statistical and Machine-Learning Methods

3.1 Cox proportional hazards regression

The Cox proportional hazards model expresses the hazard for an individual with covariate vector $\mathbf{x}$ as

$$h(t | \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}), \quad (2)$$

where $h_0(t)$ is the baseline hazard and $\boldsymbol{\beta}$ is the vector of regression coefficients.

The exponential term represents the relative hazard associated with the covariate vector.

The model was fitted using an L_2 -penalized Cox formulation.

The penalization parameter was set to

$$\lambda = 0.01, \quad (3)$$

with an L_1 ratio of zero.

Penalization was used to regularize the fitted coefficients.

The fitted model was evaluated on both the training and held-out test cohorts.

Patient-level risk scores were obtained from the model's predicted partial hazards.

Five-year survival probabilities were obtained by evaluating the fitted survival function at 60 months.

The Cox model was implemented using the `lifelines` Python library.

3.2 Random Survival Forest

Random Survival Forests extend random forest methods to censored time-to-event data (Ishwaran et al., 2008).

The model constructs an ensemble of survival trees using bootstrap samples of the training observations.

At each candidate split, a subset of predictors is considered.

The ensemble combines information across individual trees to estimate survival-related quantities.

Unlike a conventional Cox model, a Random Survival Forest does not require a linear relationship between the predictors and the log hazard.

This flexibility allows the model to represent nonlinear patterns and interactions that may be difficult to specify in advance.

However, increased flexibility can also increase the risk of overfitting, particularly when the available training sample is limited.

3.3 Random Survival Forest tuning

An internal training-validation split was created within the 1,583-patient training cohort.

The internal fitting subset contained 1,266 patients, while the validation subset contained 317 patients.

The split was stratified by event status and used a random seed of 42.

The initial Random Survival Forest contained 200 trees, with `max_features` set to `sqrt`.

A limited hyperparameter search examined minimum leaf sizes of 5, 10, and 15 and minimum split sizes of 10 and 20.

Validation concordance was used to compare the candidate settings.

The final selected configuration used 200 trees, a minimum leaf size of 10, and a minimum split size of 10.

The model was subsequently refitted using the complete 1,583-patient training cohort.

Table 3: Final Random Survival Forest configuration.

Hyperparameter	Final setting
Number of trees	200
Minimum samples per split	10
Minimum samples per leaf	10
Maximum features	Square root
Random seed	42
Parallel processing	Enabled

The Random Survival Forest was implemented using the `scikit-survival` Python library (Pölsterl, 2020).

3.4 Discrimination

Model discrimination was assessed using Harrell’s concordance index (Harrell et al., 1982).

The concordance index measures the extent to which predicted risk scores correctly rank patients according to their observed survival outcomes among comparable patient pairs.

Conceptually, the concordance index can be written as

$$C = \frac{N_{\text{concordant}} + 0.5N_{\text{tied}}}{N_{\text{comparable}}}. \quad (4)$$

A value of 0.5 is commonly interpreted as discrimination comparable to random ranking, while a value of 1 indicates perfect concordance among the evaluated comparable pairs.

Censoring affects which patient pairs can be used in the calculation.

For both models, the concordance index was calculated using the observed follow-up times, event indicators, and model-generated risk scores.

Training concordance was reported to provide context for model fitting, while test concordance was used for the primary comparison of predictive discrimination.

3.5 Five-year prediction accuracy

The five-year Brier score was used to evaluate the accuracy of predicted survival probabilities.

For an uncensored binary outcome, the Brier score is the mean squared difference between a predicted probability and the observed outcome.

Survival data require additional consideration because the survival status at a particular time may be unknown for patients censored before that time.

The inverse probability of censoring weighted Brier score addresses this problem by weighting

evaluable observations according to the estimated censoring distribution (Graf et al., 1999).

Let $\hat{S}(t | \mathbf{x}_i)$ denote the predicted survival probability for patient i at time t , and let $\hat{G}(t)$ denote the estimated probability of remaining uncensored through time t .

The IPCW Brier score is

$$BS(t) = \frac{1}{n} \sum_{i=1}^n \left[\frac{I(T_i \leq t, \delta_i = 1)}{\hat{G}(T_i^-)} \hat{S}(t | \mathbf{x}_i)^2 + \frac{I(T_i > t)}{\hat{G}(t)} (1 - \hat{S}(t | \mathbf{x}_i))^2 \right]. \quad (5)$$

The primary evaluation time was

$$t = 60 \text{ months}. \quad (6)$$

The censoring distribution was estimated using the training cohort.

Both models were evaluated using the same training and test cohorts and the same prediction horizon.

Lower Brier scores indicate lower prediction error.

3.6 Kaplan–Meier baseline

A Kaplan–Meier model was fitted to the training cohort to provide a reference prediction.

The training-cohort Kaplan–Meier estimate of five-year survival was 0.7789.

This probability was assigned to every test patient, regardless of their clinical characteristics.

The resulting predictions were evaluated using the same five-year IPCW Brier score.

The baseline provides a reference for assessing whether individualized model predictions reduce error relative to assigning the same survival probability to all patients.

The Kaplan–Meier baseline does not produce individualized risk rankings and was therefore not included in the concordance comparison.

3.7 Calibration

Calibration was evaluated at 60 months by comparing predicted survival probabilities with observed Kaplan–Meier survival estimates.

For each model, test patients were divided into five approximately equal-sized groups according to their predicted five-year survival probabilities.

Group 1 contained patients with the lowest predicted survival, while Group 5 contained those

with the highest predicted survival.

Within each group, the mean predicted survival probability was calculated.

A Kaplan–Meier estimator was then fitted using the observed survival times and event indicators for patients in that group.

The estimated Kaplan–Meier survival probability at 60 months was used as the observed group-level survival estimate.

A model is well calibrated at the group level when the mean predicted survival probability is close to the corresponding observed estimate.

The groups were formed separately for each model. Consequently, groups with the same numerical label did not necessarily contain identical patients.

This analysis provides a descriptive assessment of grouped calibration rather than a complete evaluation of individual-level calibration.

3.8 Bootstrap uncertainty analysis

A paired bootstrap procedure was used to estimate uncertainty in the differences between the two models.

The held-out test cohort contained 397 patients.

For each bootstrap iteration, 397 patient indices were sampled with replacement from the test cohort.

The same sampled indices were applied to the Cox and Random Survival Forest predictions and outcomes.

This paired procedure preserved the dependence between the two models’ predictions for the same patients.

A total of 2,000 bootstrap iterations were performed using a fixed random seed of 42.

For discrimination, the bootstrap statistic was

$$\Delta_C = C_{\text{RSF}} - C_{\text{Cox}}. \quad (7)$$

For five-year prediction error, the bootstrap statistic was

$$\Delta_{BS} = BS_{\text{Cox}}(60) - BS_{\text{RSF}}(60). \quad (8)$$

Both differences were defined so that a positive value indicated better observed performance for the Random Survival Forest.

The 2.5th and 97.5th percentiles of the bootstrap distributions were used to construct 95%

percentile confidence intervals.

The fitted models and training cohort were held fixed during this procedure.

Therefore, these confidence intervals quantify uncertainty associated with resampling the test patients, rather than the full variability that would arise from repeating model development on different training cohorts.

4 Results

4.1 Study cohort and outcome observations

The final modeling cohort included 1,980 patients.

Of these, 1,583 were allocated to the training cohort and 397 to the held-out test cohort.

The training cohort contained 915 observed events, while the test cohort contained 229 observed events.

For the five-year evaluation in the test cohort, 84 patients experienced an observed death by 60 months.

A further 302 patients were observed beyond 60 months, and 11 patients were censored by 60 months.

These three categories accounted for all 397 test patients.

Table 4: Five-year outcome ascertainment in the test cohort.

Outcome category	Patients
Observed death by 60 months	84
Observed beyond 60 months	302
Censored by 60 months	11
Total	397

The distinction between observed events and censoring was retained throughout model fitting and evaluation.

4.2 Random Survival Forest validation

The internal validation analysis examined six candidate combinations of minimum leaf size and minimum split size.

The candidate with a minimum leaf size of 10 and a minimum split size of 10 achieved a validation concordance index of approximately 0.6741.

The corresponding training concordance index within the internal fitting subset was approximately 0.7319.

Other tested configurations produced validation concordance indices between approximately 0.6723 and 0.6741.

The differences among these candidate configurations were small.

The selected Random Survival Forest was subsequently refitted using the complete training cohort.

4.3 Predictive discrimination

The penalized Cox model achieved a training concordance index of 0.6596 and a test concordance index of 0.6548.

The Random Survival Forest achieved a training concordance index of 0.7297 and a test concordance index of 0.6912.

Table 5: Discrimination of the two survival prediction models.

Model	Training C-index	Test C-index
Cox proportional hazards	0.6596	0.6548
Random Survival Forest	0.7297	0.6912

The observed difference in test concordance was

$$\Delta_C = 0.691247 - 0.654836 = 0.036411. \quad (9)$$

The Random Survival Forest therefore demonstrated higher discrimination than the Cox model in the held-out test cohort.

The Random Survival Forest also showed a larger difference between training and test concordance.

This observation is consistent with greater flexibility during model fitting, although it does not by itself establish the extent of overfitting.

4.4 Five-year prediction accuracy

The five-year IPCW Brier score was 0.1698 for the Kaplan–Meier baseline.

The penalized Cox model achieved a five-year Brier score of 0.1570.

The Random Survival Forest achieved a five-year Brier score of 0.1451.

Table 6: Five-year IPCW Brier scores on the held-out test cohort.

Model	Five-year IPCW Brier score
Kaplan–Meier baseline	0.1698
Cox proportional hazards	0.1570
Random Survival Forest	0.1451

Both individualized prediction models achieved lower five-year Brier scores than the Kaplan–Meier baseline.

The observed difference between Cox regression and the Random Survival Forest was

$$\begin{aligned}
\Delta_{BS} &= BS_{\text{Cox}}(60) - BS_{\text{RSF}}(60) \\
&= 0.156995 - 0.145089 \\
&= 0.011906.
\end{aligned}
\tag{10}$$

Thus, the Random Survival Forest demonstrated lower five-year prediction error in the held-out test cohort.

4.5 Five-year calibration

Calibration was evaluated using five prediction-based groups for each model.

Each model’s test patients were divided into groups containing approximately 79 or 80 individuals.

Table 7 presents the mean predicted five-year survival probability and the corresponding observed Kaplan–Meier estimate.

Table 7: Five-year grouped calibration for both models.

Model	Group	Patients	Predicted	Observed
Cox	1	80	0.5796	0.5947
	2	79	0.7427	0.8222
	3	79	0.8078	0.7536
	4	79	0.8533	0.8839
	5	80	0.9067	0.8743
RSF	1	80	0.5542	0.5640
	2	79	0.7177	0.6425
	3	79	0.8015	0.8318
	4	79	0.8681	0.9487
	5	80	0.9290	0.9375

For the Cox model, the mean predicted survival probability in Group 1 was 0.5796, compared with an observed estimate of 0.5947.

In Group 2, observed survival was higher than predicted: 0.8222 versus 0.7427.

In Group 3, observed survival was lower than predicted: 0.7536 versus 0.8078.

The remaining Cox groups also showed differences between predicted and observed survival, although these differences were smaller in magnitude.

For the Random Survival Forest, Group 1 had a mean predicted survival probability of 0.5542 and an observed survival estimate of 0.5640.

Group 5 had a mean predicted survival probability of 0.9290 and an observed estimate of 0.9375.

Larger discrepancies were observed in intermediate groups.

In Group 2, predicted survival was 0.7177, compared with observed survival of 0.6425.

In Group 4, predicted survival was 0.8681, compared with observed survival of 0.9487.

These findings indicate that neither model achieved perfect grouped calibration across the full range of predicted survival probabilities.

4.6 Calibration visualization

Figure 1 displays the combined five-year calibration results.

The diagonal line represents perfect calibration.

Points above the diagonal indicate that observed survival was higher than predicted survival.

Points below the diagonal indicate that observed survival was lower than predicted survival.

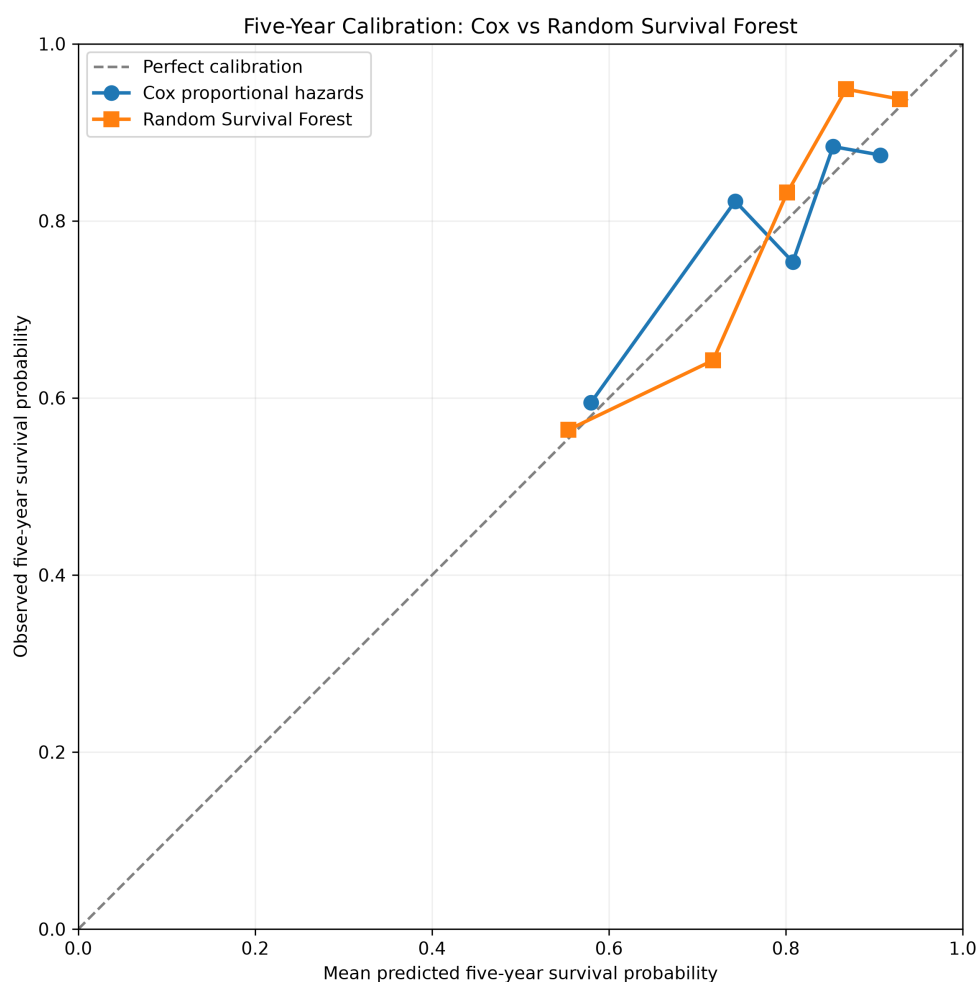


Figure 1: Five-year calibration of Cox proportional hazards regression and Random Survival Forest. Each point represents a prediction-based patient group. The diagonal line represents perfect calibration.

The plot provides a visual comparison of grouped calibration but does not establish the statistical significance of differences in calibration between the two models.

4.7 Bootstrap comparison of discrimination

The paired bootstrap procedure produced 2,000 successful resamples for the concordance comparison.

The observed test concordance difference was 0.0364.

The 95% percentile bootstrap confidence interval was

$$95\% CI_{\Delta_C} = [0.0089, 0.0655]. \quad (11)$$

The interval was entirely above zero.

Under the paired test-set bootstrap procedure, this result supports a positive difference in discrimination for the Random Survival Forest.

4.8 Bootstrap comparison of prediction error

The paired bootstrap analysis of five-year Brier score differences also produced 2,000 resamples.

The observed difference was 0.0119, defined as the Cox Brier score minus the Random Survival Forest Brier score.

The 95% percentile bootstrap confidence interval was

$$95\% CI_{\Delta_{BS}} = [0.0023, 0.0219]. \quad (12)$$

This interval was also entirely above zero.

The result supports a lower five-year prediction error for the Random Survival Forest under the specified bootstrap procedure.

Table 8: Paired bootstrap comparison of model performance.

Metric	Difference	Lower 95% CI	Upper 95% CI
Test C-index: RSF minus Cox	0.0364	0.0089	0.0655
Five-year Brier: Cox minus RSF	0.0119	0.0023	0.0219

The bootstrap intervals reflect variation in the sampled test patients while holding the trained models fixed.

They should not be interpreted as confidence intervals incorporating all uncertainty associated with data preprocessing, hyperparameter selection, model fitting, or external population differences.

5 Discussion

5.1 Principal findings

This study compared penalized Cox proportional hazards regression and Random Survival Forest for overall survival prediction using the same clinical predictor set and held-out METABRIC test cohort.

Three principal observations emerged.

First, the Random Survival Forest achieved higher test-set discrimination than the Cox model.

Second, the Random Survival Forest achieved lower five-year IPCW Brier score.

Third, both models showed discrepancies between predicted and observed survival probabilities in some calibration groups.

The paired bootstrap confidence intervals for the discrimination and Brier score differences were entirely above zero in the direction favoring the Random Survival Forest.

Together, these results indicate that the Random Survival Forest provided more accurate risk ranking and five-year probabilistic prediction than the penalized Cox model in this particular held-out cohort.

However, the results do not establish that the Random Survival Forest would perform better across all breast cancer populations or clinical settings.

5.2 Interpretation of discrimination

The concordance index evaluates whether patients assigned higher predicted risk tend to experience events earlier than patients assigned lower predicted risk, among comparable pairs.

The Cox model achieved a test concordance index of approximately 0.655.

The Random Survival Forest achieved approximately 0.691.

The observed difference suggests that the Random Survival Forest captured additional information relevant to the ordering of patient survival outcomes.

One possible explanation is that the flexible tree-based structure represented nonlinear predictor relationships or interactions that were not explicitly included in the Cox model.

For example, the association between a clinical variable and survival may differ across subgroups defined by other characteristics.

A conventional Cox model with only main effects does not automatically represent such interactions.

The Random Survival Forest can represent some of these patterns through recursive partitioning.

Nevertheless, the present analysis did not directly test which nonlinear relationships or interac-

tions contributed to the performance difference.

The explanation therefore remains a plausible modeling interpretation rather than an empirically established mechanism.

5.3 Interpretation of five-year prediction accuracy

The Brier score evaluates the accuracy of predicted survival probabilities at a specified time horizon.

Unlike the concordance index, it considers the numerical values of predicted probabilities, not only their ordering.

The Random Survival Forest achieved a five-year IPCW Brier score of 0.1451.

The corresponding Cox score was 0.1570.

Both models achieved lower prediction error than the Kaplan–Meier baseline score of 0.1698.

The baseline comparison indicates that using the selected clinical predictors provided additional predictive information beyond assigning the same training-cohort survival probability to every test patient.

The lower Random Survival Forest Brier score suggests that its predicted five-year survival probabilities were, on average, more accurate under the IPCW evaluation procedure.

However, the Brier score combines aspects of discrimination and calibration.

It should not be interpreted as a standalone measure of calibration.

5.4 Interpretation of calibration

Calibration is particularly important when a model produces absolute survival probabilities.

A predicted five-year survival probability of 0.80 should, in an appropriately defined group of similar predictions, correspond to an observed survival frequency near 80%.

The grouped calibration analysis showed that both models produced predictions reasonably close to observed survival in some patient groups.

However, meaningful differences were observed in others.

For the Random Survival Forest, the predicted and observed estimates were close in the lowest and highest predicted-survival groups.

Larger discrepancies occurred in two intermediate groups.

The Cox model also showed differences between predicted and observed survival in intermediate groups.

These results illustrate why model discrimination and prediction accuracy should be considered

alongside calibration.

A model can rank patients effectively while producing imperfect absolute probability estimates.

The present calibration analysis was descriptive.

The five groups contained approximately 80 patients each, and the Kaplan–Meier estimates were subject to sampling uncertainty.

The analysis did not establish a statistically significant difference in calibration between the models.

5.5 Model complexity and interpretability

Cox regression provides regression coefficients that can be expressed as hazard ratios under its modeling assumptions.

This structure supports interpretation of the direction and magnitude of associations between covariates and the modeled hazard.

Random Survival Forests do not produce an equivalent single set of regression coefficients.

Their predictions are derived from an ensemble of survival trees.

Although the Random Survival Forest achieved better observed performance on the two primary metrics in this study, predictive performance and interpretability are distinct model properties.

The present study did not evaluate feature importance, partial dependence, individual explanation methods, or clinical interpretability of the Random Survival Forest.

Consequently, the findings should not be interpreted as establishing that one model is preferable for every clinical or research purpose.

5.6 Implications for survival modeling

The results illustrate the value of comparing statistical survival models and machine-learning survival models under a common evaluation framework.

A model comparison can be misleading when different predictors, patient samples, preprocessing procedures, or prediction horizons are used.

In this study, the same processed clinical features and held-out test cohort were used for both models.

The primary evaluation metrics were also applied consistently.

This design supports a more direct comparison of the predictive behavior of the two algorithms.

The findings also demonstrate that a model comparison should include more than a single discrimination metric.

The combination of concordance, IPCW Brier score, grouped calibration, and paired bootstrap uncertainty provides a broader description of predictive performance.

6 Limitations

6.1 Single-cohort retrospective evaluation

The study used one retrospective breast cancer research cohort.

Although the held-out test set was not used for final model fitting, it was drawn from the same underlying data source as the training cohort.

The results therefore represent internal test-set performance.

They do not establish how either model would perform in a different hospital, geographic region, treatment period, or patient population.

Independent external validation is necessary before considering clinical deployment.

6.2 Limited predictor set

The analysis used seven clinical predictors.

The models did not include genomic features, molecular subtypes, treatment information, detailed comorbidity measures, or other potentially relevant prognostic variables.

The observed predictive performance is therefore conditional on the information available in the selected clinical feature set.

Adding further predictors could change the performance of either model.

However, the present study did not investigate whether additional variables would improve prediction.

6.3 Single training–test partition

The primary performance estimates were obtained from one fixed training–test split.

Different partitions could produce different fitted models and different performance estimates.

The paired bootstrap analysis resampled the existing test patients without refitting the models.

It therefore does not quantify the full uncertainty associated with alternative training samples, preprocessing estimates, or hyperparameter selection.

Repeated nested cross-validation or bootstrap model development could provide additional information about model stability.

6.4 Limited hyperparameter search

The Random Survival Forest was tuned using a limited set of candidate minimum leaf sizes and minimum split sizes.

Other hyperparameters, including the number of trees and number of candidate features at each split, were not extensively optimized.

The selected configuration should therefore be interpreted as a reasonable evaluated configuration, not as a globally optimal model.

Likewise, the Cox model used a fixed penalization parameter of 0.01.

A more extensive and consistently nested hyperparameter selection procedure for both models could change their relative performance.

6.5 Grouped calibration

Calibration was assessed using five groups formed according to each model's predicted survival probabilities.

This approach provides an interpretable descriptive summary but can obscure variation within groups.

The grouping also depends on the distribution of predictions in the particular test cohort.

The present analysis did not calculate a smooth calibration curve, calibration intercept, calibration slope, or formal confidence intervals for the difference in calibration between models.

Accordingly, the calibration findings should be interpreted as exploratory.

6.6 Censoring assumptions

The IPCW Brier score depends on estimation of the censoring distribution.

Its interpretation requires appropriate assumptions concerning the censoring mechanism and the adequacy of the censoring model.

The present analysis used a Kaplan–Meier-based estimate of the censoring distribution from the training cohort.

If censoring is strongly associated with clinical predictors or other unmeasured factors, a marginal censoring model may be insufficient.

The study did not investigate alternative covariate-dependent censoring models.

6.7 Clinical applicability

The analysis evaluated statistical prediction performance, not clinical decision-making.

No decision-curve analysis, prospective evaluation, or clinical utility assessment was performed.

The models were not evaluated for fairness across demographic or clinical subgroups.

The predicted survival probabilities should not be used to guide individual patient care without

appropriate external validation, clinical review, and evaluation of potential harms.

7 Future Research Directions

Several extensions could strengthen the present analysis.

First, independent external validation could determine whether the observed performance differences generalize beyond the METABRIC cohort.

Second, repeated nested cross-validation could be used to estimate variability associated with model selection and fitting.

Third, additional survival models could be evaluated, including gradient-boosted survival methods, survival support vector machines, and neural-network survival models.

Fourth, feature importance and model explanation methods could be used to investigate which clinical variables contribute most strongly to the Random Survival Forest predictions.

Such analyses would require careful distinction between predictive importance and causal effects.

Fifth, calibration could be examined at additional prediction horizons, including shorter-term and longer-term survival.

Finally, future work could evaluate subgroup performance and the potential clinical utility of individualized survival predictions.

These extensions would help establish whether the observed predictive advantages are stable, reproducible, and relevant to practical clinical applications.

8 Conclusion

This independent computational study compared penalized Cox proportional hazards regression and Random Survival Forest for overall survival prediction using clinical variables from the METABRIC breast cancer cohort.

The analysis included 1,583 training patients and 397 held-out test patients.

The Random Survival Forest achieved a test concordance index of 0.6912, compared with 0.6548 for the Cox model.

The corresponding five-year IPCW Brier scores were 0.1451 and 0.1570, respectively.

Paired bootstrap confidence intervals supported positive differences in discrimination and five-year prediction accuracy for the Random Survival Forest under the specified test-set resampling procedure.

Both models showed imperfect calibration in some prediction-based patient groups.

Overall, the findings demonstrate that a flexible ensemble survival model can achieve higher observed predictive performance than a conventional penalized Cox model when both are trained

and evaluated using the same clinical predictors and patient cohort.

The findings should be interpreted as an internal retrospective model comparison, not as evidence of clinical readiness.

External validation, additional model stability analyses, and more detailed calibration assessment are required before the predictions can be considered for clinical use.

Declarations

Author contributions

Monica Chaganti conducted the independent computational analysis, including data preparation, survival modeling, performance evaluation, calibration analysis, and bootstrap comparison.

Funding

No external funding is reported for this independent computational study.

Competing interests

The author should confirm the absence or presence of any relevant competing interests before manuscript submission.

Data availability

The analysis used the METABRIC breast cancer research dataset. The final manuscript should identify the exact dataset release, repository, and access date used in the analysis.

Code availability

The analysis was implemented in Python using a reproducible notebook-based workflow. The final manuscript should provide a verified public repository link if the analysis code is released.

Ethics statement

This study involved secondary computational analysis of an existing research dataset. The applicable data-use conditions and any institutional ethics requirements should be verified before submission.

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A Reproducibility and Computational Workflow

The study was organized as a notebook-based computational workflow.

The principal stages were:

1. Construction and validation of the survival analysis cohort.
2. Exploratory survival analysis using Kaplan–Meier methods.
3. Investigation of clinical predictor associations using Cox regression.
4. Preparation of a common training and test feature representation.
5. Development and evaluation of the penalized Cox prediction model.
6. Development, internal validation, and evaluation of the Random Survival Forest.
7. Comparison of model discrimination, five-year prediction error, and calibration.
8. Paired bootstrap estimation of uncertainty in the performance differences.

The primary processed data files were:

```
prepared_data/  
  X_train_processed.csv  
  X_test_processed.csv  
  y_train.csv  
  y_test.csv  
  clinical_preprocessor.joblib
```

The predictive models and evaluation results were saved separately to support reproducibility.

```
models/  
  random_survival_forest.joblib  
  
results/  
  cox_predictive_model.joblib  
  cox_predictive_performance.csv  
  cox_test_survival_predictions.csv  
  cox_five_year_brier_comparison.csv  
  cox_five_year_calibration.csv  
  rsf_metrics.csv  
  rsf_test_predictions.csv  
  rsf_calibration_5yr.csv
```

```

final_model_comparison.csv
final_model_calibration_5yr.csv
final_bootstrap_comparison.csv
final_research_summary.csv

```

The exact software environment and package versions should be recorded alongside the final public code release.

B Detailed Model Comparison

Table 9: Summary of the final predictive model comparison.

Measure	Cox	RSF
Training patients	1,583	1,583
Test patients	397	397
Processed predictors	12	12
Training C-index	0.6596	0.7297
Test C-index	0.6548	0.6912
Five-year IPCW Brier score	0.1570	0.1451
Prediction horizon	60 months	60 months

B.1 Interpretation of the reported metrics

The concordance index quantifies the agreement between predicted risk rankings and observed survival ordering among comparable pairs.

It does not directly measure the numerical accuracy of individual survival probabilities.

The five-year Brier score evaluates probability prediction error at 60 months, with weighting to account for censoring.

The calibration analysis compares predicted and observed survival within patient groups.

These measures describe complementary aspects of model performance and should be interpreted together.

B.2 Scope of the bootstrap intervals

The bootstrap confidence intervals were generated from paired resampling of the held-out test cohort.

The original training cohort and fitted models remained fixed.

Accordingly, the intervals do not represent uncertainty from the complete model development process.

A future study could repeat preprocessing, hyperparameter tuning, and model fitting within each resampling iteration to investigate development-stage variability.